

MAT1100 Homework 1

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Question 1

(a) *Proof.* Consider $\sigma, \tau \in S_4$ then

$$(\sigma \circ \tau) \cdot (i, j) = (\sigma \circ \tau(i), \sigma \circ \tau(j)) = \sigma \cdot (\tau(i), \tau(j)) = \sigma \cdot (\tau \cdot (i, j))$$

and

$$\text{Id} \cdot (i, j) = (\text{Id}(i), \text{Id}(j)) = (i, j)$$

for all $(i, j) \in X$ therefore this operation is an action of S_4 on X .

To determine the orbits of this action, first take $d = (i, i) \in X$ for $i \in \{1, 2, 3, 4\}$. Then for all $\sigma \in S_4$

$$\sigma \cdot (i, i) = (\sigma(i), \sigma(i))$$

therefore $\mathcal{O}_d \subseteq \{(i, i) \in X : i \in \{1, 2, 3, 4\}\}$. Now take given $j \in \{1, 2, 3, 4\}$, take $\sigma = (i, j)$

$$\sigma \cdot (i, i) = (j, j)$$

therefore

$$\mathcal{O}_d = \{(i, i) \in X : i \in \{1, 2, 3, 4\}\}$$

now consider $i \neq j \in \{1, 2, 3, 4\}$. Given $\sigma \in S_4$, it is bijective so $\sigma(i) \neq \sigma(j)$. Thus if $a = (i, j)$

$$\mathcal{O}_a \subseteq \{(i, j) \in X : i \neq j \in \{1, 2, 3, 4\}\}$$

and moreover given $k \neq \ell \in \{1, 2, 3, 4\}$ the permutation $\sigma = (i, k)(j, \ell)$ acts such that

$$\sigma \cdot (i, j) = (k, \ell)$$

hence

$$\mathcal{O}_a = \{(i, j) \in X : i \neq j \in \{1, 2, 3, 4\}\}$$

Clearly $\mathcal{O}_d \sqcup \mathcal{O}_a = X$ hence these are the distinct orbits. □

(b) *Proof.* Let $G = S_4$ and G_x denote the stabilizer of $x \in X$. First consider $(1, 1) \in X$. $\sigma \in G_{(1,1)} \iff \sigma \cdot (1, 1) = (1, 1) \iff \sigma(1) = 1$. Therefore

$$G_{(1,1)} = \{\sigma \in S_4 : \sigma(1) = 1\}$$

clearly any permutation in the stabilizer can freely permute $\{2, 3, 4\}$ and therefore

$$|G_{(1,1)}| = |S_3| = 6$$

Now consider $(1, 2) \in X$. $\sigma \in G_{(1,2)} \iff \sigma \cdot (1, 2) = (1, 2) \iff \sigma(1) = 1, \sigma(2) = 2$. Thus

$$G_{(1,2)} = \{\sigma \in S_4 : \sigma(1) = 1, \sigma(2) = 2\}$$

clearly any permutation in the stabilizer can freely permute $\{3, 4\}$ and therefore

$$|G_{(1,2)}| = |S_2| = 2$$

□

(c) *Proof.* Consider $\sigma \in S_4$. Let $\rho : S_4 \rightarrow \text{Sym}(X)$ be the permutation representation of the action $S_4 \curvearrowright X$.

$$\sigma \in \ker \rho \iff \forall (i, j) \in X, \sigma \cdot (i, j) = (i, j) \iff \forall i \in \{1, 2, 3, 4\}, \sigma(i) = i \iff \sigma = \text{Id}$$

hence $\ker \rho = \{\text{Id}\}$.

□